

Solution algorithm

1.Importing and installing libraries.

2.Data preparation

- *Encoding labels: 0 for fail and 1 for pass.

- *creating a 2-d array using NumPy and creating data frame using pandas library containing given test data: age, temperature, pass/fail label.

- *Separation of feature and label.

3. KNN Model And classification

- *Applying KNN classification model on prepared data by taking $k=5$.

- *N-neighbours = 5

4.Creating function for plotting datapoints and decision boundary

- *Generating mesh grid.

- *Predicting class labels for each point in mesh grid.

- *Reshaping result to create a grid of prediction.

- *Plotting decision boundary and data points and then plotting training points.

- *Plotting scatter chart.

5.Carrying out LOOCV and comparing result

- *Performing LOOCV for different k-values in range.

- *Creating KNN classifier and fitting to training data.

- *Calculating average accuracy across all folds.

- *Plotting LOOCV result.

- *Finding optimum value of K by comparing maximum accuracy correspond to k.

Possible validation Procedure

1.Cross-Validation

- * Dividing given dataset into training and validation datasets.

*Training KNN model on training set and validating performance on validation set.

2.Outliers

*An outlier is a data point that significantly deviates from rest of data in dataset. Either to remove or correct outliers.

3.Visualization

*Plotting scatter chart and visually inspecting it.

4.Parameter tuning

5.Overfitting check.